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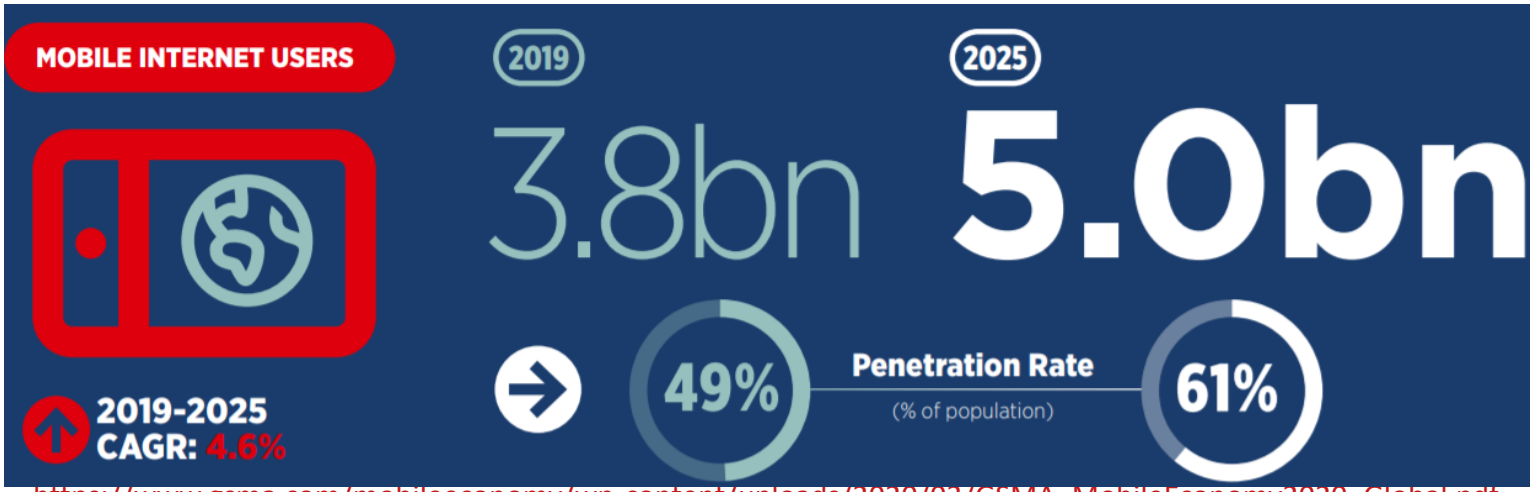
Towards Resilient Data Management for the Internet of Moving Things

Datenbanksysteme für Business, Technologie und Web
BTW 2021

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Motivation

- Mobile devices are increasingly used in IoT scenarios:
  - They have become ubiquitous.
  - Ever-growing computing capabilities.
  - Ability to generate data.



https://www.gsma.com/mobileeconomy/wp-content/uploads/2020/03/GSMA_MobileEconomy2020_Global.pdf

Problem

- Challenges:
  - Transient disconnections.
  - Mobility within the network.
- Existing fault tolerance strategies:
  - Checkpointing.
  - Replication.

- **Assumption that data sources can always be replicated.**
- **Failure to leverage offline computing capabilities.**

Solution Overview



NebulaStream

Reactive buffering-based fault tolerance strategy for mobile data sources.

At disconnection

- Assume transient disconnection.
- Remove device after timeout.

During disconnection

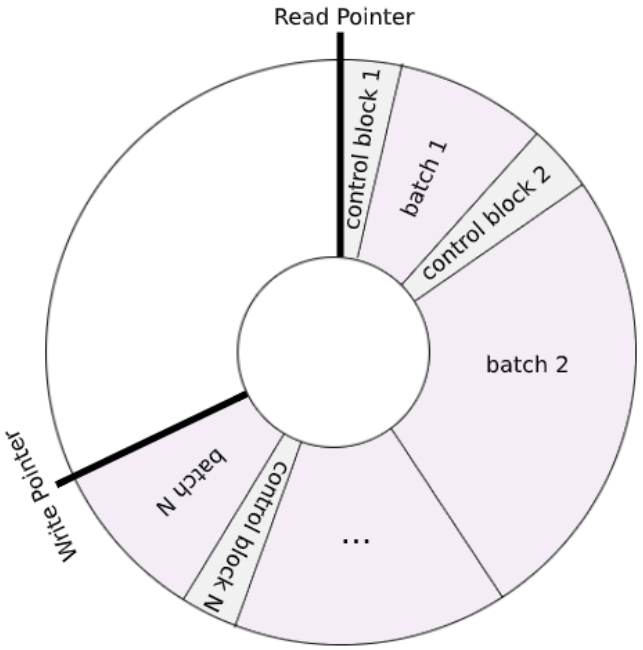
- Buffering.

At reconnection

- Selective redeployment.
- Auxiliary forwarding queries.

In-memory Buffering

- Circular buffer.
- Data arrives at the buffer after being processed offline.
- Batches are preceded by a control block indicating:
  - Query identifier.
  - Size of each tuple.
  - Number of tuples.
- Buffer size can be customized based on device storage capacity, data generation rate and expected disconnection time.



In-memory Buffering: Replacement Strategies

- **FIFO**: Starting from Read Pointer we remove as many batches of results as we need to make space for the new data.
- **Query-Aware FIFO**: Attempts to always keep some results of each query.
  - If possible, it replaces the oldest results of the same query.
  - Otherwise, it replaces the oldest results of queries with the higher number of batches currently stored in the buffer.
  - Might suffer from segmentation → Shifting data.

Reconnection

1. Source notifies the coordinator of its return:
   - Indicates the queries it was involved in.
   - Transmits its position by announcing which devices are in its neighbourhood.

2. Coordinator determines for each query if the current position of the source affects its execution.
   - If not → Resumption of query execution.
   - Otherwise → query redeployment.

Reconnection: After Redeployment

Three scenarios based on the **operators assigned to the source**:

1 Same

Execution can continue as normal

2 Superset

The data in the buffer is not fully processed.

→

New operators need to be applied before sending it.

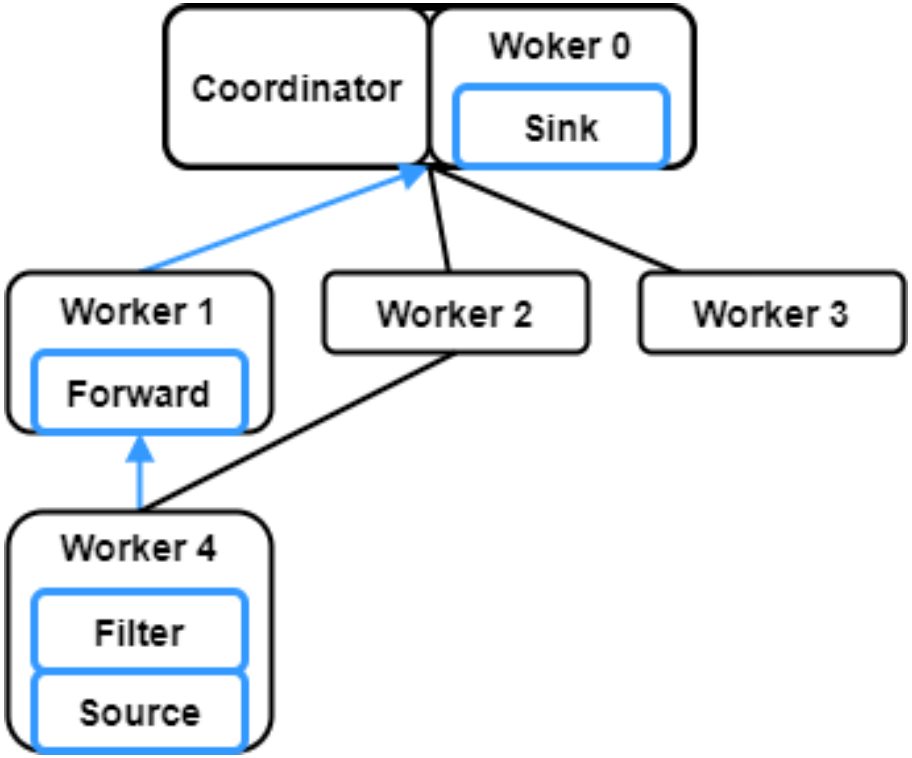
3 Subset

If sent directly, some operators would be applied to the data twice.

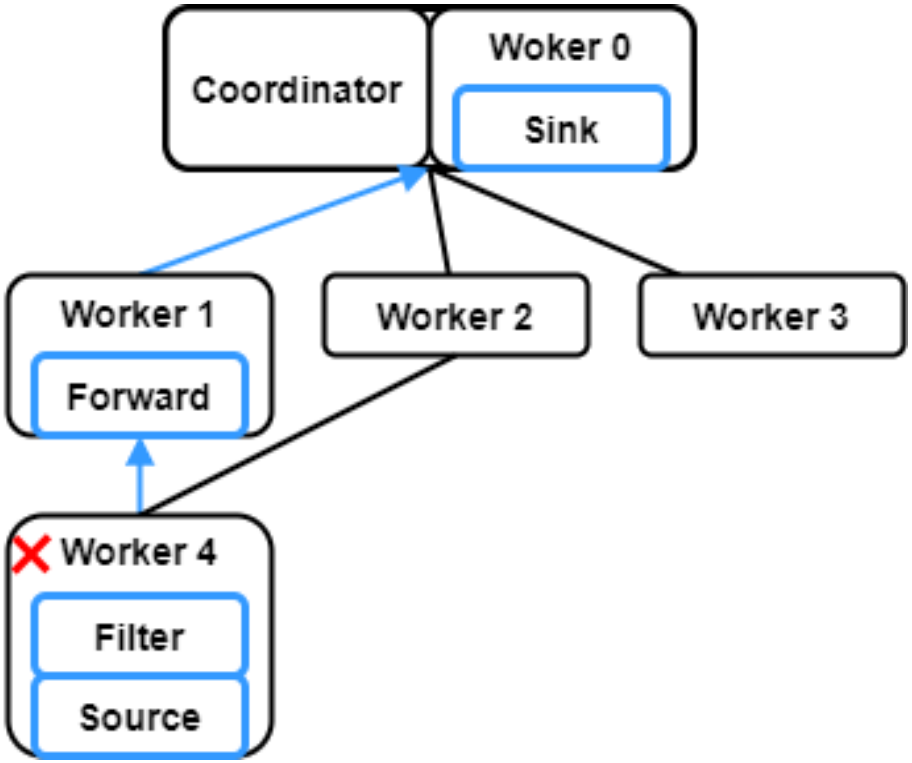
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Forwarding queries.

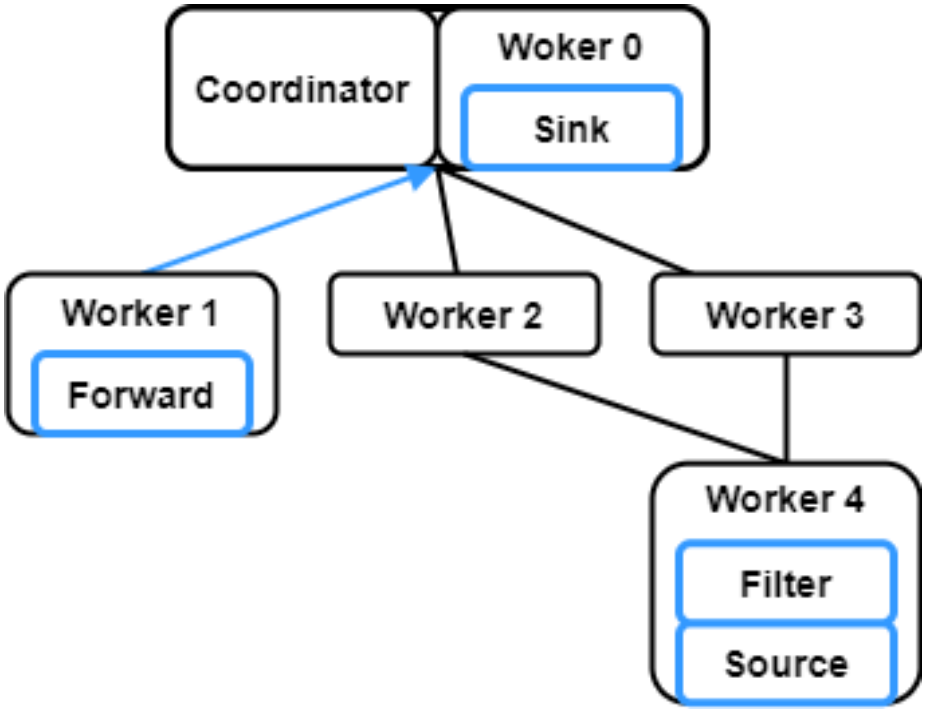
Reconnection: Forwarding Queries



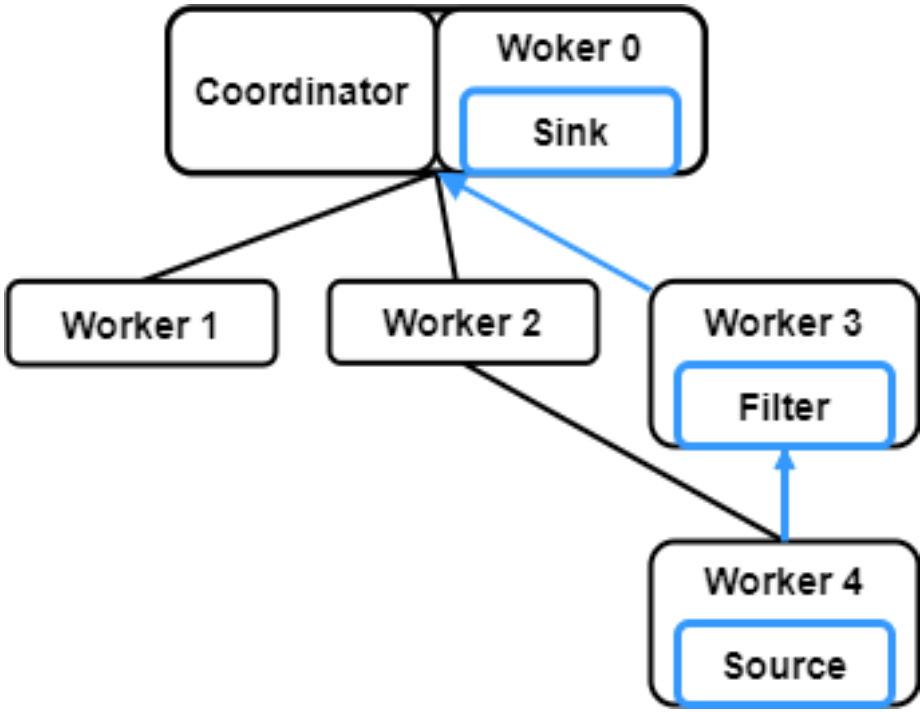
Reconnection: Forwarding Queries



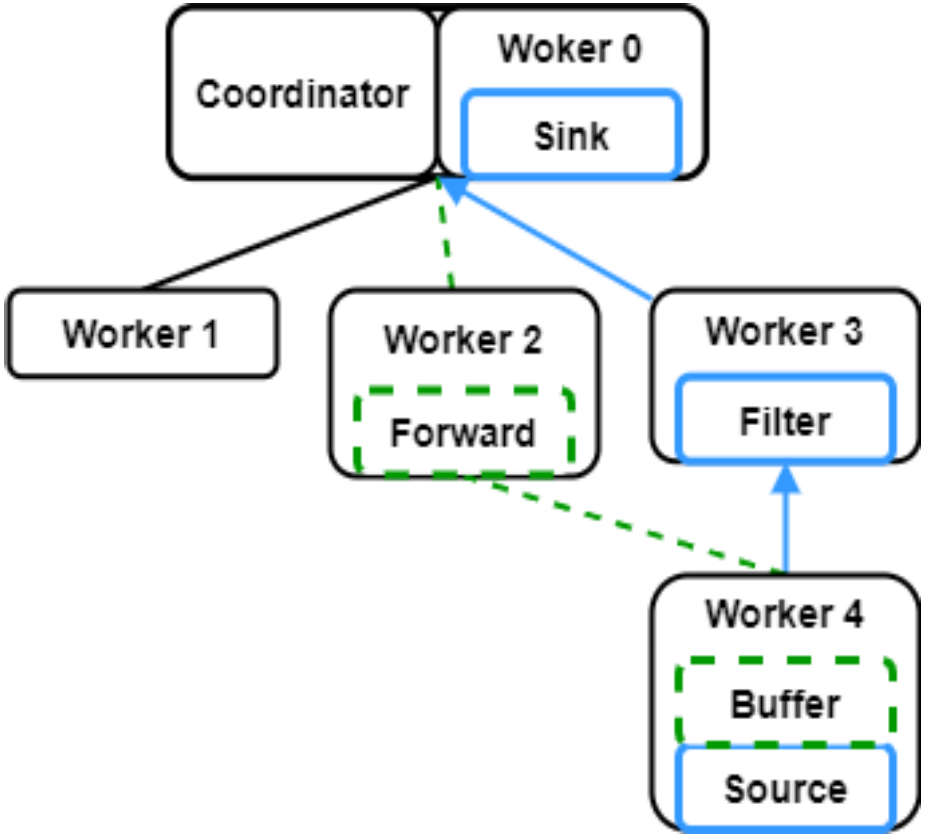
Reconnection: Forwarding Queries



Reconnection: Forwarding Queries



Reconnection: Forwarding Queries

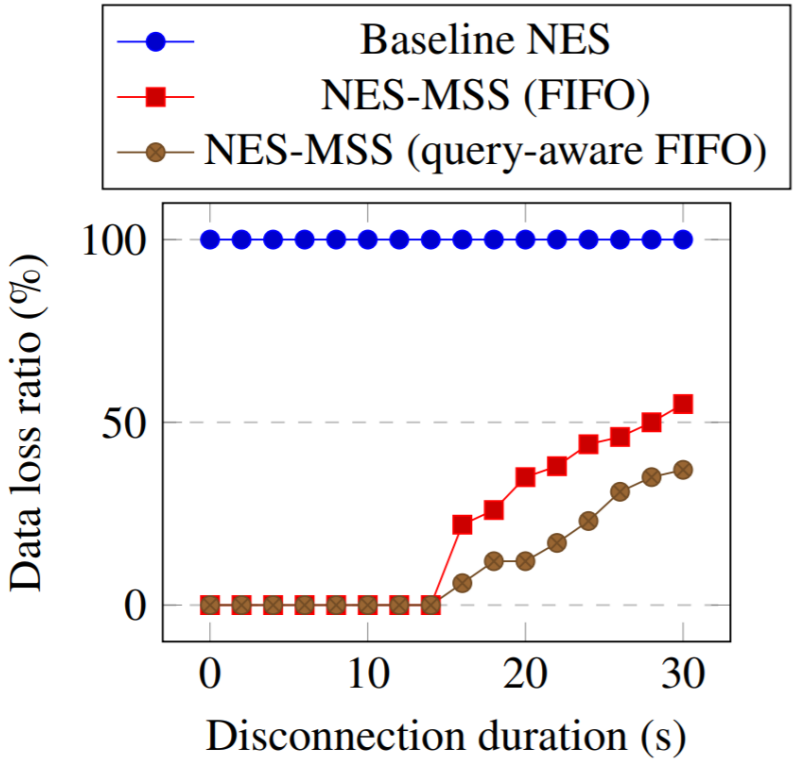


Evaluation: Setup

- **Queries**: Two queries. Each with two filter operators.
  - CSV → 3.33 MBps.
  - Binary generator → 488 KBps.
- **Baseline**: NebulaStream: remove nodes on disconnection. If a node reconnects to the network, it is treated as an entirely new node and queries are deployed from scratch.

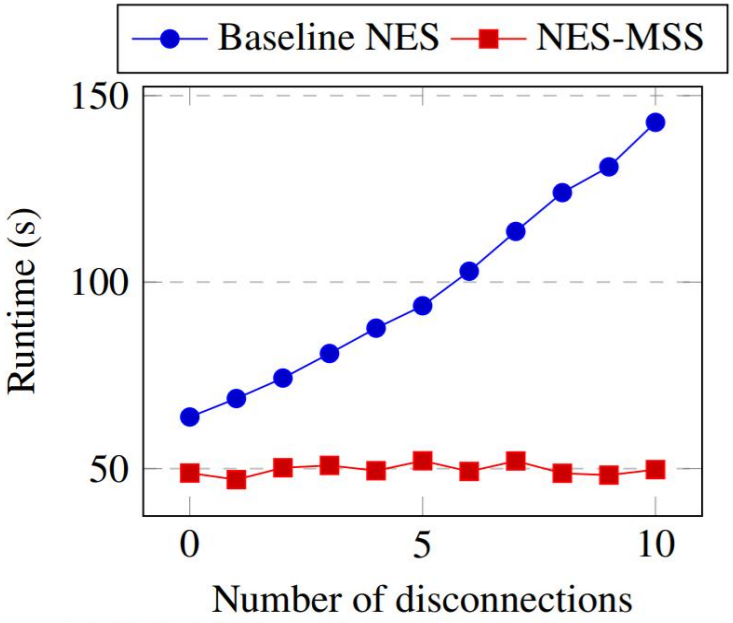
All experiments were run locally in a machine with a 10th generation Intel Core i7 processor and 16GB RAM running Ubuntu 20.04.

Evaluation: Data Loss Ratio

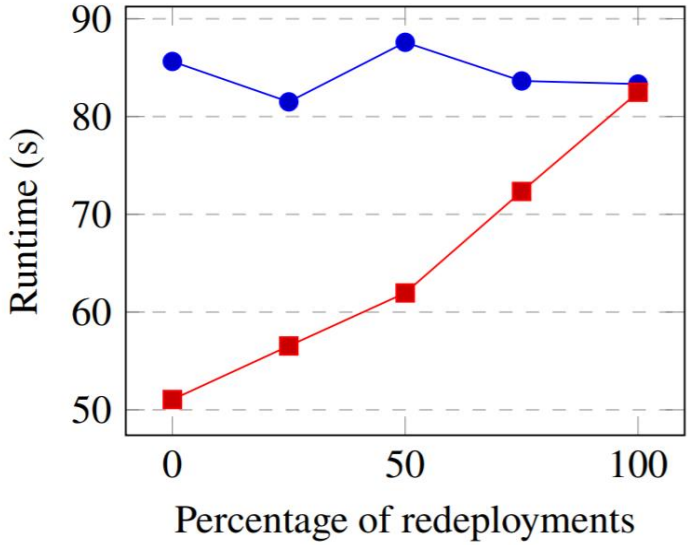


Our circular buffer reduces the amount of discarded data during transient disconnections.

Evaluation: Query Runtime



(a) NES-MSS avoids query redeployment.



(b) Impact of query redeployment.

Selective redeployment reduces the impact of transient disconnections on the query runtime.

Conclusions

- Mitigate the impact of transient disconnections.
- Exploit offline computing capabilities:
  - data processed and stored in circular buffer.
- Handle mobility:
  - selective query redeployment and forwarding queries.

- Compared to baseline NebulaStream:
  - Reduced data loss.
  - Reduced query runtime.

} When transient disconnections occur.